

Project Checkpoint + Hard Portal Cuts and a Momentum-Frequency Retarded Kernel

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral;
see RECOVERY_PROVENANCE.md.

Executive result

The project checkpoint separates the formal chronometry and protected-QCD phenomenology into defensible paper units. The hard portal contribution is 7.98260×10^{-4} T and the LPM contribution is 3.60256×10^{-4} T, giving $\Gamma_{H,\text{total}}/T = 1.1585159 \times 10^{-3}$ and $\Gamma_H/\Gamma_R = 7.8514 \times 10^6$. The momentum-frequency kernel is a causal KMS-complete near-shell reconstruction anchored to that exact on-shell rate.

Reported benchmark values

- metric_1: 0.00079826
- metric_2: 0.000360256
- metric_3: 0.0011585159
- metric_4: 116083.0
- metric_5: 7851400.0